DATA ANALYTICS CAPSTONE PROJECT GUIDE

INDIVIDUAL PROJECTS | 8 WEEKS | PRODUCTION-BASED DELIVERY

This handbook is for every student completing the two-month capstone. Each student will plan, build, test, document and present one independent analytics product. Students may work in the same project category, but the final projects must be different in problem, dataset, user flow or decision supported.

IMPORTANT - THIS IS NOT A GROUP PROJECT

One student = one project, one GitHub repository, one Power BI report, one working data pipeline and one final viva. You may discuss ideas with classmates, but the implementation, evidence and explanation submitted for assessment must be your own.

Programme coverage

Excel	SQL	Python	Machine Learning	Power BI	AI Tools	Automation	Deployment
Validation	Data model	Pipeline & EDA	Model & evaluation	Business report	Useful AI feature	n8n/ workflow	Working or reproducible

Class size: 17 students | Categories: 4 | Initial category capacity: 4 students each | One additional slot allotted by the instructor

How to use this guide

Read Sections 1 to 4 before proposing an idea. Use the category pages as a reference while selecting a problem. Complete the Week 1 proposal before starting full development. Keep the weekly evidence pages updated throughout the eight weeks.

Section	What it helps you do
1. Project rules	Understand the individual-project structure and allocation process.
2. Production scope	Understand what the final product must contain.
3. Four categories	Choose a suitable problem and understand what strong projects look like.
4. Approval process	Prepare three ideas and get one approved.
5. Eight-week plan	Know what must be demonstrated every week.
6. Tool guidance	Choose a sensible technical stack.
7. AI usage	Use AI responsibly and keep evidence.
8. Final submission	Prepare deployment, documentation, presentation and viva evidence.

THE THREE QUESTIONS BEHIND EVERY APPROVED PROJECT

1. What real problem are you solving?
2. What decision will your product improve?
3. What action happens after the analysis or prediction?

1. Individual project structure and category allocation

Every student will complete an independent capstone. The four categories are used only to distribute topics and provide direction. They are not project teams.

Rule	What it means
Independent ownership	Each student is responsible for the complete project from problem definition to deployment and final explanation.
Category capacity	Four initial seats are available in each category. Since there are 17 students, one additional seat will be allotted by the instructor after reviewing preferences and project diversity.
Different projects	Students in the same category cannot submit the same problem with the same dataset and same solution flow.

Rule	What it means
Preference system	Each student submits a first and second category preference. Allocation depends on seat availability and idea quality.
Week 1 approval	Full development starts only after the instructor approves the problem, data and scope.
Individual evidence	Every student maintains their own repository, work log, weekly demo evidence and final viva preparation.

How category allocation will work

1. Submit your first and second category preferences.
2. Prepare three possible project ideas from the category you prefer.
3. The instructor checks whether the category still has a seat and whether the ideas are sufficiently different from other students.
4. If the first category is full or your idea duplicates another project, you may be moved to your second category or asked to change the problem.
5. The additional seventeenth seat will be assigned by the instructor. It may be added to one category only when the proposed project remains clearly different.

SAME CATEGORY DOES NOT MEAN SAME PROJECT

For example, four students may work under Customer and Revenue Intelligence. One may build a churn intervention system, another may study discount effectiveness, another may build lead scoring, and another may develop product recommendations. Their datasets, users, outputs and deployment flows must be different.

2. What the final capstone must become

This is not a dashboard-only assignment, a notebook-only exercise or a model-training task. The expected result is a small but complete analytics product that another person can open, understand and use.

Layer	Minimum evidence expected
Business problem	A named user or stakeholder, current pain point, decision to improve and measurable success criteria.
Excel	Validation workbook, business mapping, reconciliation, manual input template or data dictionary.
SQL	Schema or database design, joins, reusable analytical queries, aggregations and data-quality checks.
Python	Ingestion or cleaning pipeline, EDA, feature engineering, reusable functions and error handling.
Machine learning	Clear task, baseline, correct split, suitable metrics, error analysis, limitations and saved predictions/model.

Layer	Minimum evidence expected
Power BI	Overview page, diagnostic page and operational/action page with tested measures and filters.
AI feature	Useful explanation, extraction, summarisation, recommendation, search or assistant function that is validated.
Automation	Scheduled run, alert, email, task creation, workflow or report delivery using n8n or another suitable tool.
Deployment	Working link where possible, or a documented setup that another person can reproduce.
GitHub	Repository, README, issues, setup steps, screenshots, architecture, testing evidence and final release.

Expected product flow

BUSINESS PROBLEM -> DATA -> QUALITY CHECKS -> SQL/PYTHON -> ML/ANALYTICS -> POWER BI/APP -> AUTOMATED ACTION -> DEPLOYMENT

Every tool must have a reason. Do not add technology only to make the project look large.

Dataset restriction

Do not reuse the exact datasets already used in class assignments, including Online Retail, Instacart, AI4I 2020 Predictive Maintenance or the QuickBite classroom dataset. You may work in a similar domain, but the final dataset and project problem must be different.

3. Category A - Customer and Revenue Intelligence

This category deals with customers, sales, revenue, profit, products, pricing, discounts, campaigns and purchasing behaviour. The purpose is to help a business understand where revenue is coming from, where money is being lost and what action should be taken next.

Who may use the product?

Sales managers, marketing teams, customer-success teams, pricing teams, product managers, revenue heads, regional managers or small-business owners.

Questions this category can answer

- Which customers are valuable, inactive or likely to leave?
- Which products create revenue but weak profit?
- Are discounts improving demand or only reducing margin?
- Which leads or customers should be contacted first?
- What product or action should be recommended to a customer?
- Which campaign or segment gives the best return?

Strong project directions

Project direction	What a strong version should do
Customer churn and retention system	Predict customers who may become inactive. Add revenue-at-risk, reasons for risk, a priority list and a recommended retention action. A weekly automation can send the highest-risk customers to the responsible person.
Customer lifetime value and segmentation	Estimate customer value and group customers by behaviour. The product should recommend different treatment for loyal, new, inactive, discount-dependent and high-potential customers.
Pricing and discount effectiveness	Study how discount level affects sales, profit and margin. Build a recommendation for where discount should be reduced, maintained or targeted only to selected customers.
Product recommendation or next-best-product	Use purchase history to recommend relevant products. Show recommendation logic, expected opportunity and a simple interface where a user can enter a customer or product.
Campaign performance and lead scoring	Predict which leads are likely to convert, compare campaign return and create a priority queue for the sales team.

A strong example

WEAK: “Customer churn prediction.”

STRONG: “Build a customer-retention intelligence system for a subscription business. Predict churn risk, estimate revenue at risk, explain the main reasons, recommend a retention action, show the result in Power BI and send a weekly high-risk alert.”

Data grain to define

One row may represent one customer, one order, one customer per month, one campaign interaction or one customer-product pair. The final proposal must clearly state what one row represents.

Possible ML tasks

- Churn classification, purchase prediction, customer-value regression, sales forecasting, segmentation, recommendation, campaign-response prediction or lead scoring.

Power BI expectations

- Executive revenue overview, customer or segment page, churn/risk page, product or campaign page and an operational action list.

AI and automation ideas

- AI: customer summaries, segment explanations, next-best-action suggestions, campaign-message drafts or natural-language search.
- Automation: weekly high-risk alert, daily lead-scoring workflow, scheduled campaign report or assignment of top opportunities.

Ideas that are too weak

- A simple sales dashboard.
- Basic segmentation with only charts.
- Sales prediction with no decision or action.
- A churn model that cannot explain risk.
- A recommendation system copied from a tutorial without evaluation.

Questions you must be ready to answer

- Who will use the product?
- What revenue or customer decision changes?
- What does one row represent?
- What happens after the prediction?
- How will you measure business value?
- Why is this more than a dashboard?

4. Category B - Operations and Supply Chain Intelligence

This category focuses on how a company runs day to day. It may cover delivery, inventory, warehouses, transportation, production, service delays, machine maintenance, staffing or capacity. The goal is to reduce delay, waste, downtime, cost and service failure.

Who may use the product?

Operations managers, warehouse teams, delivery managers, plant managers, maintenance teams, service managers, capacity planners or regional operations heads.

Questions this category can answer

- Which orders or cases are likely to breach SLA?
- Which products may become out of stock or remain overstocked?
- Where is the operational bottleneck?
- Which machine or asset needs attention?
- How much demand will arrive, and what resources are required?
- What should the operations team act on today?

Strong project directions

Project direction	What a strong version should do
Delivery-delay prediction and intervention	Predict whether an order will breach SLA or how late it may be. Create a risk score, reason, priority queue and action such as escalation, rerouting, reassignment or customer notification.
Inventory stockout and overstock intelligence	Forecast demand, identify stockout and slow-moving risk, recommend reorder quantity and show warehouse or product-level action lists.
Predictive maintenance	Predict failure risk or remaining useful life. Convert probability into maintenance priority and compare the cost of missed failure against unnecessary inspection.
Demand and capacity planning	Forecast demand by time and location, then translate the forecast into staff, vehicle, counter, bed, warehouse or service-capacity requirements.
SLA and process bottleneck analysis	Analyse a multi-stage process, detect where delay is created, predict breach risk and create an escalation queue for unresolved cases.

A strong example

WEAK: “Delivery dashboard.”

STRONG: “Build an operations control system that predicts late deliveries before SLA breach, explains the main risk factors, places risky orders in a priority queue and sends an alert with the recommended intervention.”

Data grain to define

One row may represent one delivery, one order, one machine reading, one product per warehouse per day, one service request or one vehicle trip. Do not join order-level and item-level data without controlling duplication.

Possible ML tasks

- Delay classification, delivery-time regression, demand forecasting, stockout prediction, failure prediction, anomaly detection, remaining-useful-life estimation or capacity forecasting.

Power BI expectations

- Operations control tower, SLA/delay page, inventory health, demand forecast, resource utilisation, root-cause page and operational action queue.

AI and automation ideas

- AI: incident summaries, root-cause explanation, maintenance-note generation or an operations assistant.
- Automation: stockout alert, high-delay notification, maintenance ticket, scheduled forecast update or daily operations report.

Ideas that are too weak

- A basic delivery or inventory dashboard.
- A demand forecast with no resource action.
- A machine-failure model with no maintenance priority.
- A delay model that uses information available only after delivery.
- An operations report without a current action queue.

Questions you must be ready to answer

- What operational failure are you preventing?
- At what time is the prediction made?
- What action follows the result?
- Which error is more costly?
- How often should the pipeline run?
- What happens if data is missing or late?

5. Category C - Risk, Fraud, Quality and Compliance

This category is about detecting suspicious, incorrect, unsafe, poor-quality or non-compliant behaviour. The goal is usually to help a review team decide what to investigate first, not to make a final accusation automatically.

Who may use the product?

Risk teams, fraud investigators, claims teams, audit teams, compliance teams, finance reviewers, quality-control teams or operational review teams.

Questions this category can answer

- Which transaction, claim, invoice or case needs review first?
- What behaviour looks unusual compared with normal history?
- Which batches, suppliers or processes have quality risk?
- How should limited reviewers be allocated?
- What threshold balances missed risk and false alarms?
- How can the system explain why a case was flagged?

Strong project directions

Project direction	What a strong version should do
Transaction fraud risk and review queue	Create a risk probability, reason, priority and action such as approve, hold, verify or escalate. Evaluate false positives and false negatives.
Insurance or claims prioritisation	Identify suspicious, duplicate or high-cost cases and create a review queue. The model supports an investigator and should not automatically reject a claim.
Quality-failure prediction	Predict defect or failure risk from production conditions, supplier information, machine settings or inspection history. Create a batch or process inspection list.
Invoice or expense anomaly detection	Detect unusual amount, duplicate invoice, vendor anomaly, weekend expense, split transaction or policy violation. Produce an anomaly score and review reason.
Compliance monitoring	Combine rule-based checks and ML to identify missing approvals, sequence violations, incomplete documentation or unusual access. Send cases for human review.

A strong example

WEAK: “Fraud prediction with 98% accuracy.”

STRONG: “Build a transaction-risk review system that produces a risk score, explains the main flags, ranks cases for investigators, compares thresholds and measures the cost of false positives and missed fraud.”

Data grain and leakage

One row may represent one transaction, claim, invoice, audit case, batch or compliance event. Separate event date, review date and final outcome date. Do not use information that becomes available only after the decision.

Possible ML tasks

- Risk classification, anomaly detection, defect prediction, cost prediction, review-priority ranking, duplicate detection or suspicious-behaviour clustering.

Evaluation expectations

- Accuracy alone is not sufficient. Explain precision, recall, false-positive cost, false-negative cost, review capacity and threshold selection.

Power BI expectations

- Risk overview, high-risk cases, pattern analysis, investigation queue, threshold analysis, false-positive/false-negative review and root-cause page.

AI and automation ideas

- AI: case summaries, risk explanations, review-note drafts, policy search or document extraction.
- Automation: high-risk notification, review-ticket creation, daily risk report or automatic case assignment.

Ideas that are too weak or unsafe

- Fraud prediction evaluated only with accuracy.
- A model trained using leakage columns.
- An anomaly chart with no review process.
- A system that automatically rejects a genuine person or claim.
- A project that cannot explain false positives.

Questions you must be ready to answer

- What exactly does risk mean?
- Who makes the final decision?
- What is the cost of each error?
- Which fields exist before review?
- How will risk be explained?
- How many cases can be reviewed per day?

6. Category D - Public Impact, Healthcare and Education

This category focuses on services used by citizens, patients, students, schools, hospitals, NGOs or government departments. The aim is to improve service quality, response time, resource allocation or early support. Projects in this category must also address privacy, fairness and responsible use.

Who may use the product?

School administrators, teachers, hospital operations teams, public-service departments, NGOs, district administrators, education officers or healthcare analysts.

Questions this category can answer

- Which students or service users may need early support?
- Where will demand or congestion increase?
- Which complaints or cases need priority?
- Which institution or location has a performance gap?
- Where should limited public resources be allocated?
- Could the model unfairly affect a person or group?

Strong project directions

Project direction	What a strong version should do
Student performance and intervention system	Identify students who may need support using attendance, assignment and performance patterns. Produce a reason, intervention list and suggested support. Do not permanently label a student.
Hospital wait-time and resource planning	Forecast patient demand or waiting time and convert the result into staffing, counter, bed or department-capacity recommendations. This is an operations project, not medical diagnosis.
Public complaint prioritisation	Predict resolution delay or urgency, identify hotspots, compare department performance and create an escalation queue for unresolved or high-priority complaints.
Healthcare service-quality analysis	Compare service quality, waiting time, satisfaction, readmission or capacity indicators to identify operational gaps. Avoid making unsupported medical claims.
Public resource allocation	Forecast demand for transport, ambulances, waste collection, school resources or public services and recommend where additional resources may be needed.

A strong example

WEAK: “Student marks dashboard.”

STRONG: “Build an early-support system that identifies students showing attendance and learning-risk patterns, explains the reason, provides a teacher intervention list, tracks outcomes in Power BI and records privacy and fairness limitations.”

Data grain to define

One row may represent one student, one patient visit, one complaint, one hospital, one school, one service request or one location per day. State whether the project works at individual, institution, location or time level.

Possible ML tasks

- Student-risk classification, wait-time regression, demand forecasting, complaint-priority classification, service-delay prediction, resource-allocation clustering or quality-risk scoring.

Responsible-use requirement

- Identify sensitive fields and explain how privacy will be protected.
- Check whether the data or model may create unfair outcomes.
- State who makes the final decision.
- Explain how the result should and should not be used.
- Use the system to support a human decision, not replace it automatically.

Power BI expectations

- Service overview, institution/location comparison, demand trend, intervention or priority page, resource-allocation page, action queue and fairness/data-quality page.

AI and automation ideas

- AI: complaint summaries, student-support suggestions, policy search, category extraction or a public-service assistant.
- Automation: complaint escalation, weekly support list, congestion alert or public-service performance summary.

Ideas that are too weak or unsafe

- A simple student or hospital dashboard.
- A disease-prediction project copied from Kaggle.
- Medical diagnosis without domain expertise.
- A public-data report with no action or deployment.
- A model that automatically denies support or service.

Questions you must be ready to answer

- Who benefits from the product?
- What service decision changes?
- Is sensitive data involved?
- Could the model unfairly affect someone?
- Who makes the final decision?
- How will service impact be measured?

7. Week 1 - project selection and approval

Do not start full development with only a project title. During Week 1, every student must research three possible ideas from the allocated or preferred category and prove that the problem, dataset and scope are workable.

What to prepare for each of the three ideas

- Working project title
- Category
- Named user or stakeholder
- Current problem and why it matters
- Decision the product will improve
- Dataset source and downloaded sample
- What one row represents
- Main business metrics
- Proposed ML task
- Proposed Power BI pages
- AI feature
- Automation action
- Deployment path
- Eight-week MVP
- Optional features
- Main technical and data risks

Five-minute approval pitch

Time	What to present
1 minute	User, current problem and why it matters.
1 minute	Decision to improve and how success will be measured.
1 minute	Dataset proof: source, sample rows, fields, size and grain.
1 minute	Expected product: ML output, Power BI pages, AI feature, automation and deployment.
1 minute	MVP, optional features and biggest risks.

Approval conditions

- The stakeholder and decision are specific.
- A real data sample is available and legal to use.
- The project is large enough for eight weeks but still finishable.
- The ML task is meaningful and does not use future information.
- Power BI supports decisions rather than only displaying charts.
- The AI feature adds useful value.
- The automation performs a real action.
- The final product can be demonstrated or reproduced.

Immediate rejection or pivot conditions

- The idea is only “analyse this dataset.”
- No clear user or decision can be named.
- No sample data has been obtained.
- The project is only a dashboard or only a notebook.
- The model uses information that will not exist when prediction is made.
- The project depends on inaccessible private data or a paid service.
- The scope contains many disconnected features but no clear MVP.
- The student cannot explain what happens after the result is produced.

PROJECT STATEMENT TEMPLATE

“I am building a [type of analytics product] for [specific user]. It will use [data] to help them [decision/action]. The system will produce [main output], display [Power BI/app result], perform [automation] and be delivered through [deployment path].”

8. Eight-week execution plan

Week	Minimum review evidence
Week 1 - Approval	Three ideas, category allocation, problem statement, stakeholder, dataset sample, grain, architecture sketch, MVP and risk list.
Week 2 - Data foundation	Data dictionary, SQL schema, ingestion method, cleaning pipeline, Excel validation and data-quality checks.
Week 3 - EDA and BI v1	EDA findings, business metrics, first Power BI pages and revised problem assumptions.
Week 4 - ML baseline	Target or task design, leakage review, baseline model, split strategy, metrics and first prediction output.
Week 5 - Model improvement	Feature engineering, model comparison, error analysis, threshold or forecast review and limitations.
Week 6 - Product integration	AI feature, app/API or user flow, n8n workflow, model-to-dashboard integration and automation evidence.
Week 7 - Production readiness	Deployment, security, refresh, QA, exception handling, user testing and documentation.
Week 8 - Final release	Working demonstration, final Power BI report, GitHub release, report, presentation and individual viva.

Weekly individual review format

1. State what you committed to in the previous review.
2. Show a live demonstration or working evidence. Screenshots alone are not enough.
3. Explain one result, one error or one test finding.
4. State the current blocker and the decision you need to make.
5. Commit to the next measurable output.

INDIVIDUAL ACCOUNTABILITY

You must understand all major layers of your own project: business problem, dataset, SQL/Python flow, ML logic, Power BI, automation and deployment. AI-generated or externally generated work that you cannot explain will not count as completed work.

9. Recommended tool paths

Choose one sensible path based on the problem. Do not use every platform only to make the architecture look impressive.

Path	Suggested stack	Best suited for
A - Python analytics product	Python + SQL/Supabase + Streamlit + Power BI + n8n	Forecasting, prediction, upload-and-score apps and fast deployment.
B - Full-stack analytics product	Lovable/React + Supabase + Python API + Power BI + n8n	Login, forms, database writes, user workflows and polished interfaces.
C - BI-first operational solution	SQL + Python pipeline + Power BI + n8n + small API/app	Dashboards, action queues, alerts and scheduled operational reporting.
D - ML demonstration	Python + MLflow + Gradio/Hugging Face Spaces + Power BI	Model-centric projects requiring an interactive demonstration.

Practical role of common tools

Tool	Use it when
GitHub	You need version history, issues, release documentation and evidence of progress.
Supabase	You need managed Postgres, authentication, storage or simple APIs.
Streamlit	You need to deploy a Python analytics or ML interface quickly.
Hugging Face Spaces / Gradio	You need to host an ML demo or interactive model interface.
Lovable	You need fast UI scaffolding. You must still understand and test the generated code.

Tool	Use it when
n8n	You need a scheduled trigger, webhook, alert, email, API call or recurring workflow.
MLflow	You need to track model parameters, metrics, artifacts and comparisons.
Power BI	You need governed business metrics, diagnostic analysis and an operational action view.

10. AI usage rules

AI assistance is allowed and encouraged for research, learning, debugging, code review, documentation drafts, test cases and UI scaffolding. It does not remove the need to understand, test and validate the work.

Allowed	Not acceptable
Research concepts and compare approaches.	Submit code or a model that you cannot explain.
Ask for explanations of errors, SQL, DAX or metrics.	Fabricate data, results, citations or user testing.
Generate alternative ideas, tests or documentation drafts.	Hide limitations or model failures.
Review non-sensitive code and improve clarity.	Upload secrets, passwords or private data.
Create UI scaffolding and then verify it.	Copy a complete solution without checking whether it works.

AI usage log

Maintain a short log containing: date, tool used, task, important prompt, output used, output rejected and how the result was validated.

11. Production-readiness checklist

☐ A new user understands the purpose within two minutes.
☐ The project has a working demo link or a documented reproducible setup.
☐ The README contains architecture, setup, environment variables, data flow and screenshots.
☐ Secrets, passwords and private keys are not committed.
☐ The data pipeline handles missing files, invalid rows, schema issues or API failure sensibly.
☐ Data-quality checks produce visible results.
☐ The ML model has a baseline, valid split, suitable metrics, error analysis and limitations.
☐ The model does not use future information unavailable at prediction time.
☐ Power BI pages have clear purposes, tested measures, useful filters and readable labels.
☐ The AI feature has validation examples and known failure cases.
☐ The automation has a clear trigger, action, error path and successful-run evidence.
☐ The deployed product uses safe demo data and does not expose sensitive information.

☐ At least three users test the product and findings are recorded.

☐ The project can be rebuilt from the repository and instructions.

12. Final submission and individual viva

The final submission should prove that the product works, that the analysis is valid and that you understand the complete solution.

Deliverable	What must be included
GitHub repository	Code, issues/work log, README, architecture, setup, screenshots and final release.
Deployment	Working URL where practical, or a reproducible deployment package.
Power BI	PBIX file and report/demo evidence.
SQL	Schema, setup scripts, transformations and data-quality queries.
Python	Organised source code or notebooks forming a reproducible pipeline.
Machine learning	Experiment results, model/output, evaluation, error analysis and limitations.
Automation	n8n workflow export or equivalent, plus successful execution evidence.
AI feature	Usage log, validation examples and known limitations.
Documentation	Final report, presentation and user guide.

Typical viva questions

- Why did you choose this problem and this user?
- What does one row in the final dataset represent?
- How did you prevent leakage or unrealistic validation?
- Why did you select these metrics?
- Where does the model or analysis fail?
- How does Power BI support a decision?
- What exactly does the automation do?
- What did AI help you with, and how did you validate it?
- How can another person reproduce or deploy the project?
- What would you improve with four more weeks?

13. Project selection worksheet

Complete this worksheet for each of your three proposed ideas.

Student name	
First category preference	
Second category preference	
Proposed project title	
User / stakeholder	
Current problem	
Decision to improve	
Why it matters	
Dataset source	
What one row represents	
Main business metrics	
Proposed ML task	
Power BI pages	
AI feature	
Automation action	
Deployment path	
Eight-week MVP	

Optional features	
Main risks	

14. Weekly individual review record

Review item	Student notes
Week	
Previous commitment	
What is working now	
Live evidence shown	
Important result or finding	
Current error / blocker	
Decision taken	
Next week commitment	
Instructor feedback	

15. Final project summary

Summary item	Final response
Final project title	
Category	
User / stakeholder	
Problem solved	
Data used	
Main model or analytical method	
Power BI output	
AI feature	
Automation	
Deployment URL / setup	

Summary item	Final response
Main result	
Known limitation	
What I would improve next	

Student declaration

I confirm that this is my individual project. I can explain the business problem, data flow, SQL/Python work, machine-learning approach, Power BI report, AI usage, automation and deployment evidence included in the submission.

Student signature	Date

Instructor approval

Status	Comments / conditions
Approved / Revise / Re-present	